Sentiment Analysis using BPE and CBOW Embeddings

1. Tokenizer Selection (BPE vs WordPiece)

In this project, Byte Pair Encoding (BPE) was used for tokenization instead of WordPiece. Since embeddings were trained from scratch (without transformers), BPE was a more practical choice.

Key reasons:

- **Simple and interpretable:** Merges frequent character pairs step-by-step.
- **Fixed vocabulary size:** 30,000 tokens ensures good coverage without being too large.
- **Handles rare words:** Subword splitting reduces OOV issues.
- **Well-suited for CBOW training:** No dependency on pretrained transformer models.

Thus, BPE provided an efficient and lightweight tokenization strategy for this embedding-based pipeline.

2. CBOW + Logistic Regression

- Accuracy: 0.66
- Macro F1-score: 0.57
- Weighted F1-score: 0.65

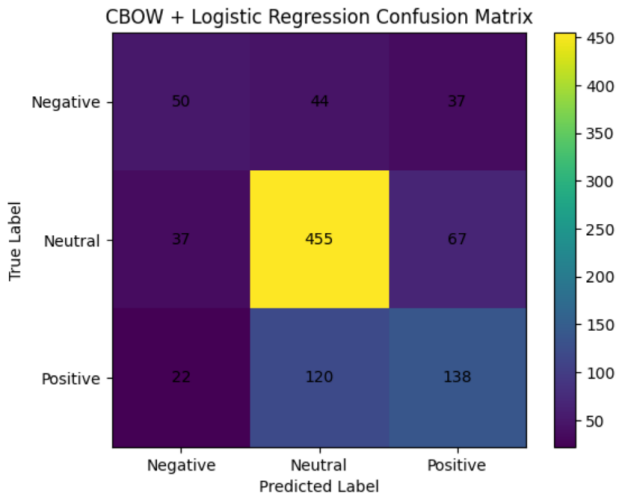


Figure 1: Confusion Matrix for CBOW + Logistic Regression

Observations:

- Neutral class predicted most accurately.
- Mild positive sentences often confused with neutral.
- Some subtle negative cases also classified as neutral.

3. VADER Baseline

- Accuracy: 0.54
- Macro F1-score: 0.49

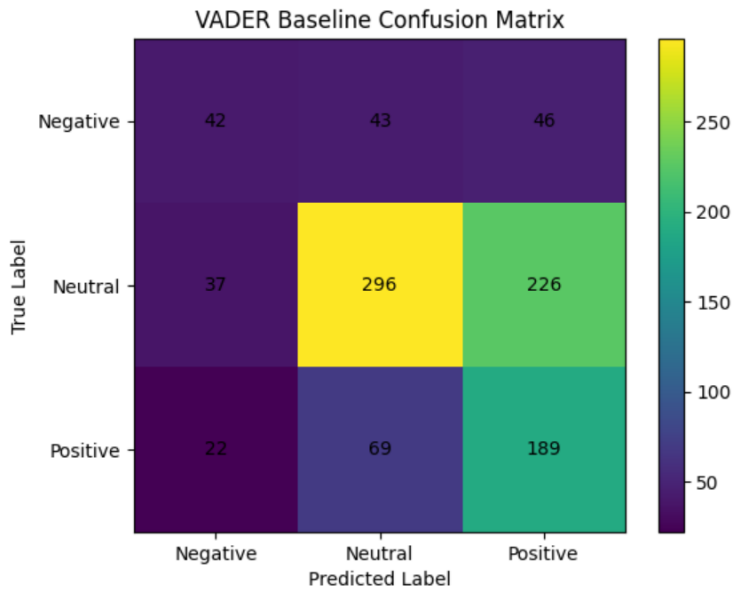


Figure 2: Confusion Matrix for VADER Baseline

Main issues with VADER:

- Large number of neutral sentences predicted as positive.
- Over-reliance on individual words like “increase” or “growth”.
- Poor adaptation to financial context.

Overall, the embedding-based model clearly outperformed the lexicon-based baseline.

4. Error Analysis

Example 1
“Result before taxes decreased to nearly EUR 14.5 mn...”

- True: Negative
- CBOW: Negative
- VADER: Neutral

The embedding model understands the financial implication of "decreased," while VADER does not capture the full economic context.

Example 2

"The shopping center will have a prime location next to the metro station."

- True: Positive
- CBOW: Positive
- VADER: Neutral

The phrase "prime location" conveys business value. CBOW captures this association through learned semantics.

Example 3

"It is in the process of taking a seat on the board."

- True: Neutral
- CBOW: Neutral
- VADER: Positive

This sentence is factual, but VADER assigns polarity based on keywords.
In summary, embeddings handled:

- Domain-specific phrasing
- Subtle financial decline
- Context-dependent sentiment

better than the rule-based system.

5. Conclusion

Key takeaways:

- Embedding-based classifier (Macro F1 = 0.57) outperformed VADER (Macro F1 = 0.49).
- Context-aware representations are more effective in financial language.
- Even simple neural models can outperform lexicon-based approaches in domain-specific tasks.